

1483. Kth Ancestor of a Tree Node

Hard

Topics

Companies

Hint

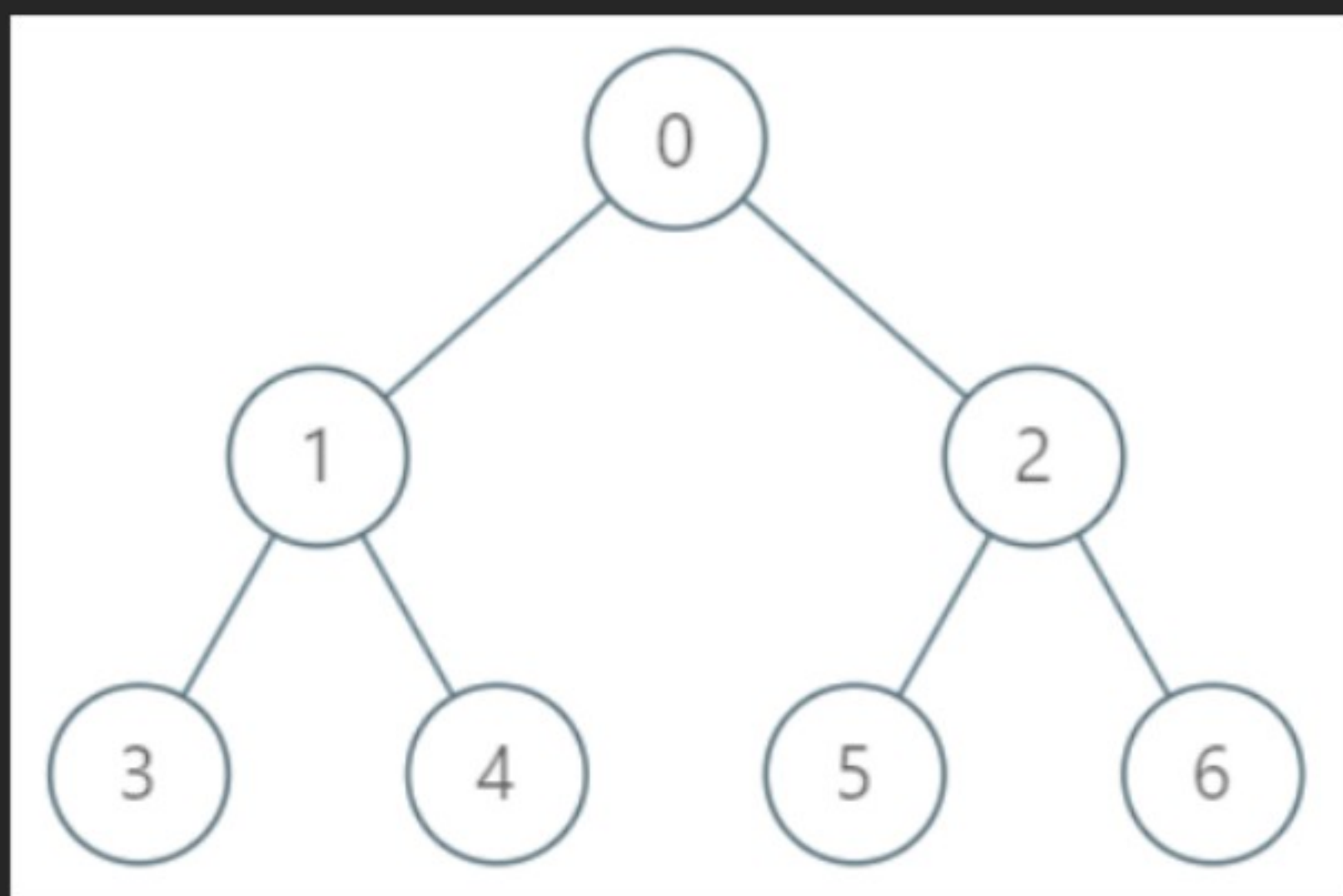
You are given a tree with n nodes numbered from 0 to $n - 1$ in the form of a parent array `parent` where `parent[i]` is the parent of i^{th} node. The root of the tree is node 0 . Find the k^{th} ancestor of a given node.

The k^{th} ancestor of a tree node is the k^{th} node in the path from that node to the root node.

Implement the `TreeAncestor` class:

- `TreeAncestor(int n, int[] parent)` Initializes the object with the number of nodes in the tree and the parent array.
- `int getKthAncestor(int node, int k)` return the k^{th} ancestor of the given node `node`. If there is no such ancestor, return `-1`.

Example 1:



Input

```
["TreeAncestor", "getKthAncestor", "getKthAncestor", "getKthAncestor"]  
[[7, [-1, 0, 0, 1, 1, 2, 2]], [3, 1], [5, 2], [6, 3]]
```

Output

```
[null, 1, 0, -1]
```

Explanation

```
TreeAncestor treeAncestor = new TreeAncestor(7, [-1, 0, 0, 1, 1, 2, 2]);  
treeAncestor.getKthAncestor(3, 1); // returns 1 which is the parent of 3  
treeAncestor.getKthAncestor(5, 2); // returns 0 which is the grandparent of 5  
treeAncestor.getKthAncestor(6, 3); // returns -1 because there is no such ancestor
```

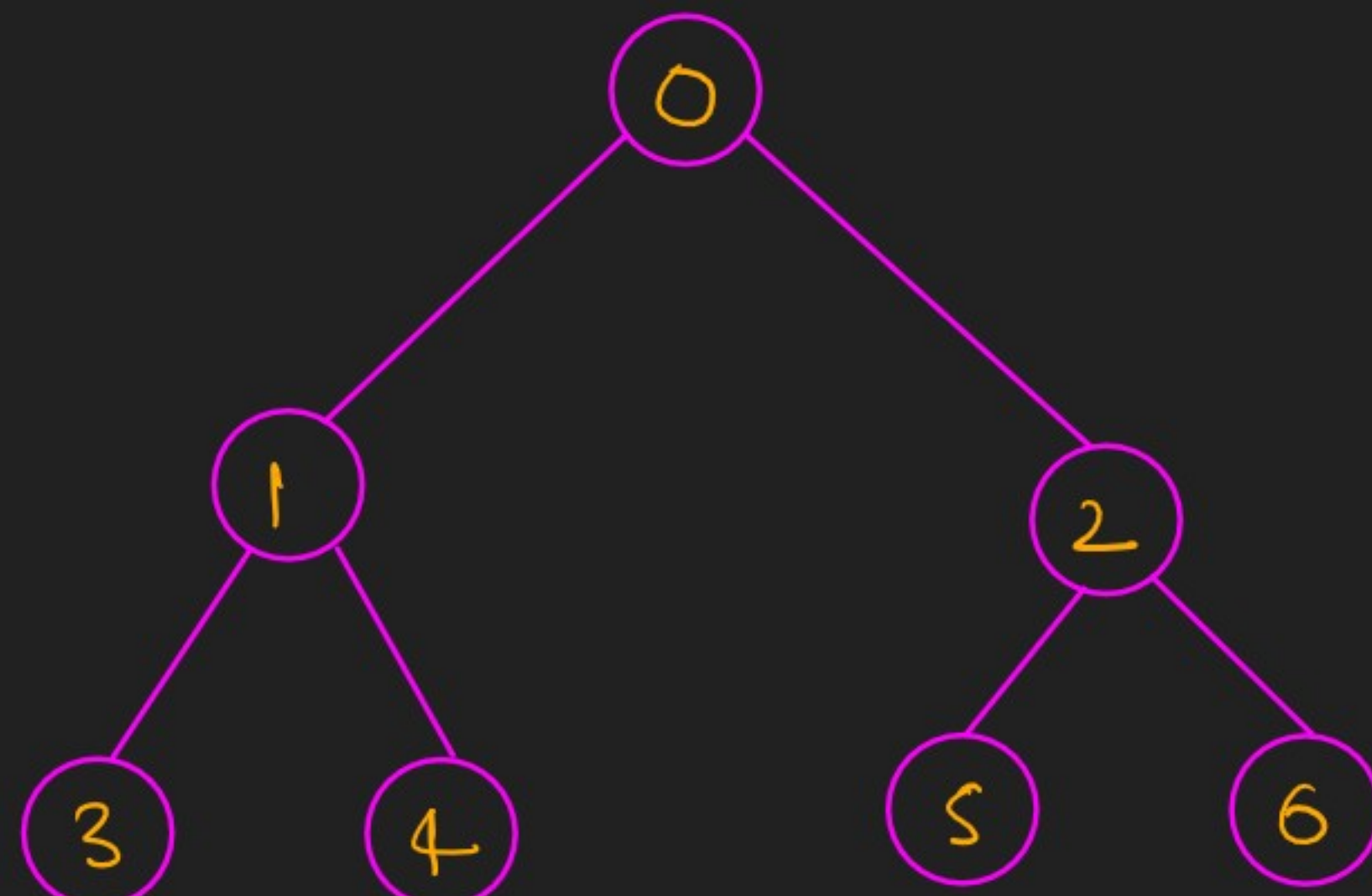
Constraints:

- $1 \leq k \leq n \leq 5 * 10^4$
- `parent.length == n`
- `parent[0] == -1`
- $0 \leq \text{parent}[i] < n$ for all $0 < i < n$
- $0 \leq \text{node} < n$
- There will be at most $5 * 10^4$ queries.

Thought Process

$$n = 7$$

$$\text{parent} = \{-1, 0, 0, 1, 1, 2, 2\}$$



$$\text{ancestor}[\text{node}][i] = ??$$

```
vector<vector<int>> ancestorTable(
    rows, vector<int>(col^n, -1)
```

$$\text{rows} = n$$

$$\text{col}^n = \log_2 7 + 1 \approx 3$$

		col ⁿ		
		0	1	2
row	0	-1		
	1	0		
	2	0		
	3	1		
	4	1		
	5	2		
	6	2		

```
for(node=0; node < n; node++){
    ancestorTable[node][0] = parent[node];
}

// Helps to find col^n - 0

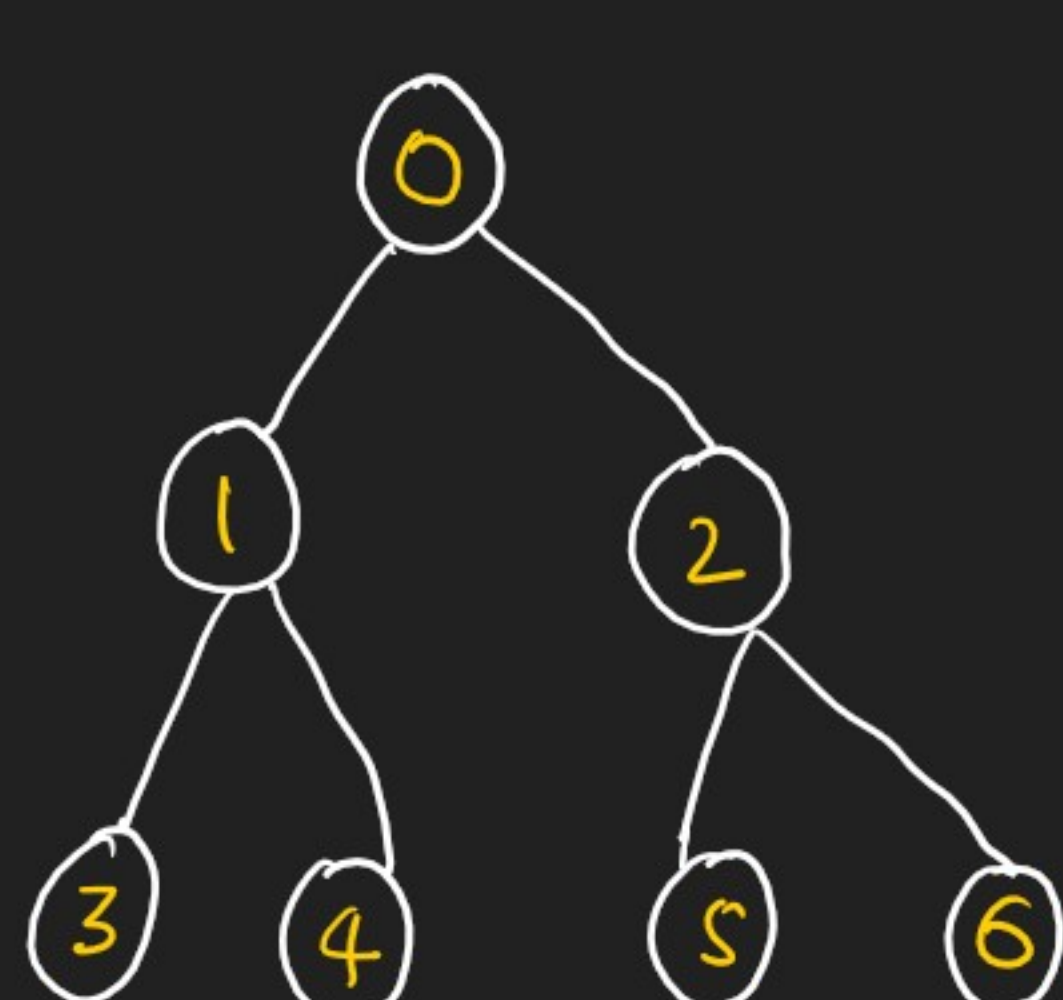
for(j=1; j <= col^n; j++){
    for(node=0; node < n; node++){
        ancestorTable[node][j] = ancestorTable
            [ancestorTable[node][j-1][j-1]];
    }
}
```

Now:

Find the k^{th} ancestor of node

int getKthAncestor(int node, int k) {

To reach k^{th} ancestor, what jumps of power of 2 will be required??



$$\text{node} = 5$$

$$k = 2$$

$$\begin{array}{c} \downarrow \text{Binary} \\ 010 = 2^1 \\ \swarrow \downarrow \searrow \\ 2^2 \quad 2^1 \quad 2^0 \end{array}$$

$$\text{ancestorTable}[5][1] = 0$$

```
for(j=0; j < cols; j++){
    if(k and (1 <= j)) { // count set bits (using left shift)
        node = ancestorTable[node][j];
        if(node == -1) {
            return -1;
        }
    }
}
```